Implement English-to-Tamil machine translation using an Encoder-Decoder model

To implement a basic Encoder-Decoder neural network for translating simple English sentences into Tamil.

Algorithm
1. Import TensorFlow and the required Keras libraries.
2. Define a small English-Tamil parallel sentence dataset inside the program.
3. Add start and end tokens to the Tamil target sentences.
4. Create separate tokenizers for English and Tamil.
5. Convert the sentences into numerical sequences.
6. Pad the source and target sequences to fixed lengths.
7. Prepare decoder input and target sequences by shifting the target sequence.
8. Create an LSTM encoder to process the English sentence.
9. Create an LSTM decoder initialized using the encoder states.
10. Add a Dense softmax layer to predict Tamil vocabulary tokens.
11. Compile and train the Encoder-Decoder model.
12. Predict a Tamil translation for a sample English sentence.

Program
```
import numpy as np
import tensorflow as tf
from tensorflow.keras import layers, Model
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
tf.keras.utils.set_random_seed(34)
# English dataset
english = [
"hello",
"thank you",
"good morning",
"how are you",
"i am fine",
"what is your name",
"good night",
"welcome",
"come here",
"go there"
]
# Tamil dataset
tamil = [
"████████ ",
"████ ██",
"█████ ████████ ",

"██ ██████ ███████ ██████████ ██████ ",
"█████ ██████ ████████████ ",
"███████ ██████ ████ ■",
"█████ █████",
"███████████████ ",
"██████ ██",
"██████ █████"
]
# Add start and end tokens
target_text = [
"<s> " + s + " </s>"
for s in tamil
]
# Tokenization
src_tok = Tokenizer(filters="")
tgt_tok = Tokenizer(
filters="",
lower=False
)
src_tok.fit_on_texts(english)
tgt_tok.fit_on_texts(target_text)
# Convert sequences
src = src_tok.texts_to_sequences(english)
tgt = tgt_tok.texts_to_sequences(target_text)
```

```
src_len = max(map(len, src))
tgt_len = max(map(len, tgt))
# Padding
encoder_data = pad_sequences(
src,
maxlen=src_len,
padding='post'
)
target_data = pad_sequences(
tgt,
maxlen=tgt_len,
padding='post'
)
decoder_input = target_data[:, :-1]
decoder_output = target_data[:, 1:]
src_vocab = len(src_tok.word_index) + 1

tgt_vocab = len(tgt_tok.word_index) + 1
# Encoder
enc_inputs = layers.Input(shape=(src_len,))
enc_emb = layers.Embedding(
src_vocab, 64
)(enc_inputs)
_, state_h, state_c = layers.LSTM(
128,
return_state=True
)(enc_emb)
# Decoder
dec_inputs = layers.Input(
shape=(tgt_len - 1,)
)
dec_emb_layer = layers.Embedding(
tgt_vocab, 64
)
dec_emb = dec_emb_layer(dec_inputs)
dec_lstm = layers.LSTM(
128,
return_sequences=True,
return_state=True
)
dec_out, _, _ = dec_lstm(
dec_emb,
initial_state=[state_h, state_c]
)
output_layer = layers.Dense(
tgt_vocab,
activation='softmax'
)
dec_out = output_layer(dec_out)
# Complete model
model = Model(
[enc_inputs, dec_inputs],
dec_out
)
# Compile

model.compile(
optimizer='adam',
loss='sparse_categorical_crossentropy',
metrics=['accuracy']
)
# Train
model.fit(
[encoder_data, decoder_input],
decoder_output[..., None],
epochs=500,
batch_size=2,
verbose=0
)
# Test translation
test_sentence = "hello"
test_src = pad_sequences(
src_tok.texts_to_sequences([test_sentence]),
```

```
maxlen=src_len,
padding='post'
)
start_id = tgt_tok.word_index["<s>"]
end_id = tgt_tok.word_index["</s>"]
generated = [start_id]
# Generate Tamil output
for _ in range(tgt_len - 1):
dec_in = pad_sequences(
[generated],
maxlen=tgt_len - 1,
padding='post'
)
probs = model.predict(
[test_src, dec_in],
verbose=0
)[0]
next_id = int(
np.argmax(probs[len(generated) - 1])
)
if next_id == 0 or next_id == end_id:
break

generated.append(next_id)
id_to_word = {
v: k for k, v in tgt_tok.word_index.items()
}
translated = " ".join(
id_to_word.get(i, "")
for i in generated[1:]
)
print("English:", test_sentence)
print("Tamil Translation:", translated)
```

Output of Program

```
English: hello
Tamil Translation: ■■■■■■■
```

Result

Thus, a basic Encoder-Decoder model was implemented for English-to-Tamil machine translation using an embedded parallel dataset. The model learned to map simple English sentences into Tamil sentences.